Problem Set 3

Applied Stats/Quant Methods 1

Due: November 13, 2025

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in `R`, please include the code you used to get your answers. Please also include the `.R` file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday November 13, 2025. No late assignments will be accepted.

In this problem set, you will run several regressions and create an add variable plot (see the lecture slides) in `R` using the `incumbents_subset.csv` dataset. Include all of your code.

Question 1

We are interested in knowing how the difference in campaign spending between incumbent and challenger affects the incumbent's vote share.

1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `difflog`.

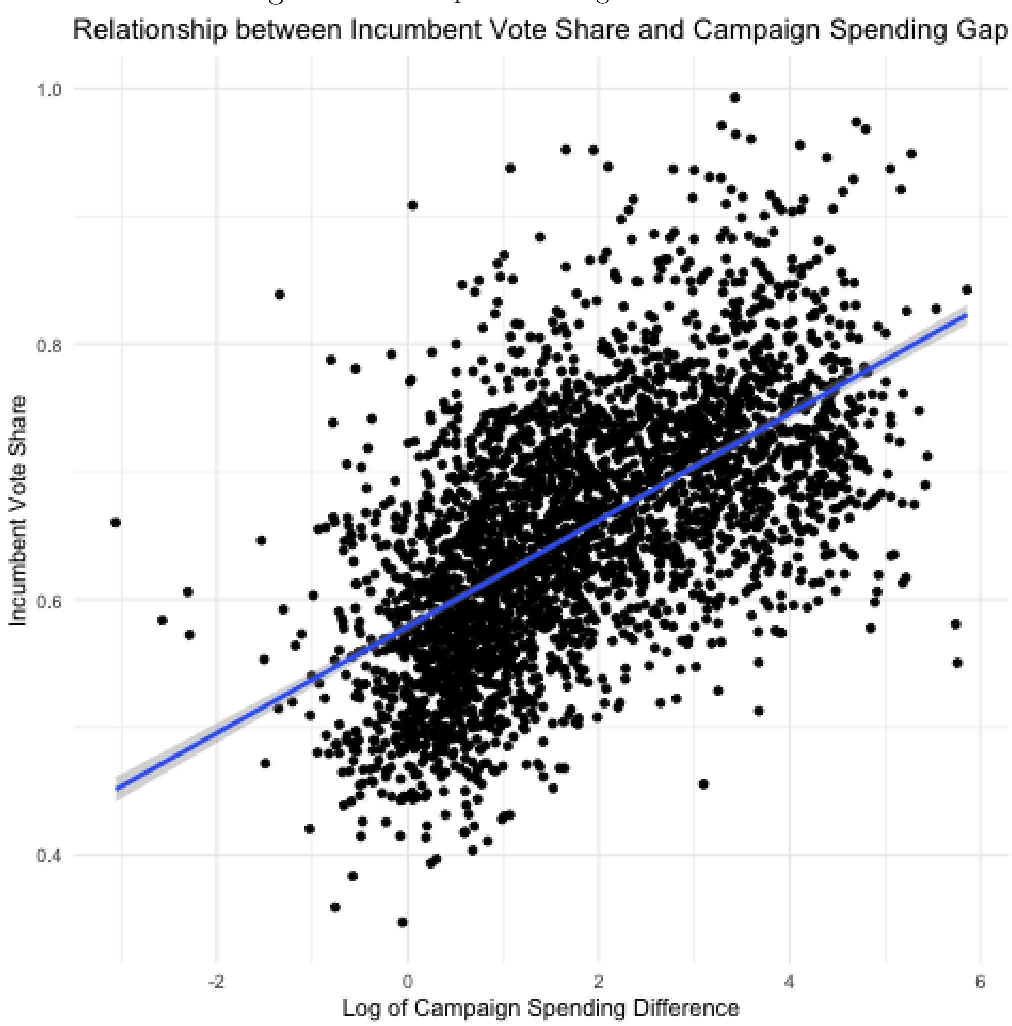
```
# read and look at data
inc.sub <- read.csv("https://raw.githubusercontent.com/ASDS-TCD/StatsI_2025/main/datasets/incumbents_subset.csv")
head(inc.sub)

# Question 1
fit1 <- lm(voteshare ~ difflog, data = inc.sub)
fit1
stargazer(fit1)
```

Table 1:

	Dependent variable:
	Incumbent Vote Share
Campaign Spending Difference	0.042***
	(0.001)
Constant	0.579***
	(0.002)
Note:	*p<0.1; **p<0.05; ***p<0.01

2. Make a scatterplot of the two variables and add the regression line.

Figure 1: scatterplot of difflog and voteshare



```r
png(file = "scatter_difflog_vs.pdf")
ggplot(inc.sub, aes(x = difflog, y = voteshare)) +
  geom_point() +
  geom_smooth(method = "lm") +  # regression line
  labs(title = "Relationship between Incumbent Vote Share and Campaign
    Spending Gap",
       x = "Log of Campaign Spending Difference",
       y = "Incumbent Vote Share") +
   theme_minimal()
dev.off()
```

3. Save the residuals of the model in a separate object.

```r
# save residuals
resid_q1 <- fit1$residuals
resid_q1
```

4. Write the prediction equation.

$$\widehat{\text{VoteShare}} = \hat{\alpha} + \hat{\beta} \times \text{DiffLog}$$
$$\widehat{\text{VoteShare}} = 0.579 + 0.042 \times \text{DiffLog}$$

The intercept $\hat{\alpha} = 0.579$, represents the expected incumbent vote share when the difference in campaign spending between incumbent and challenger is 0 (when both spend the same amount). The slope $\hat{\beta} = 0.042$ shows that for every one-unit increase in campaign spending difference, the incumbent vote share increases by 0.042, on average.

Question 2

We are interested in knowing how the difference between incumbent and challenger's spending and the vote share of the presidential candidate of the incumbent's party are related.

1. Run a regression where the outcome variable is `presvote` and the explanatory variable is `difflog`.

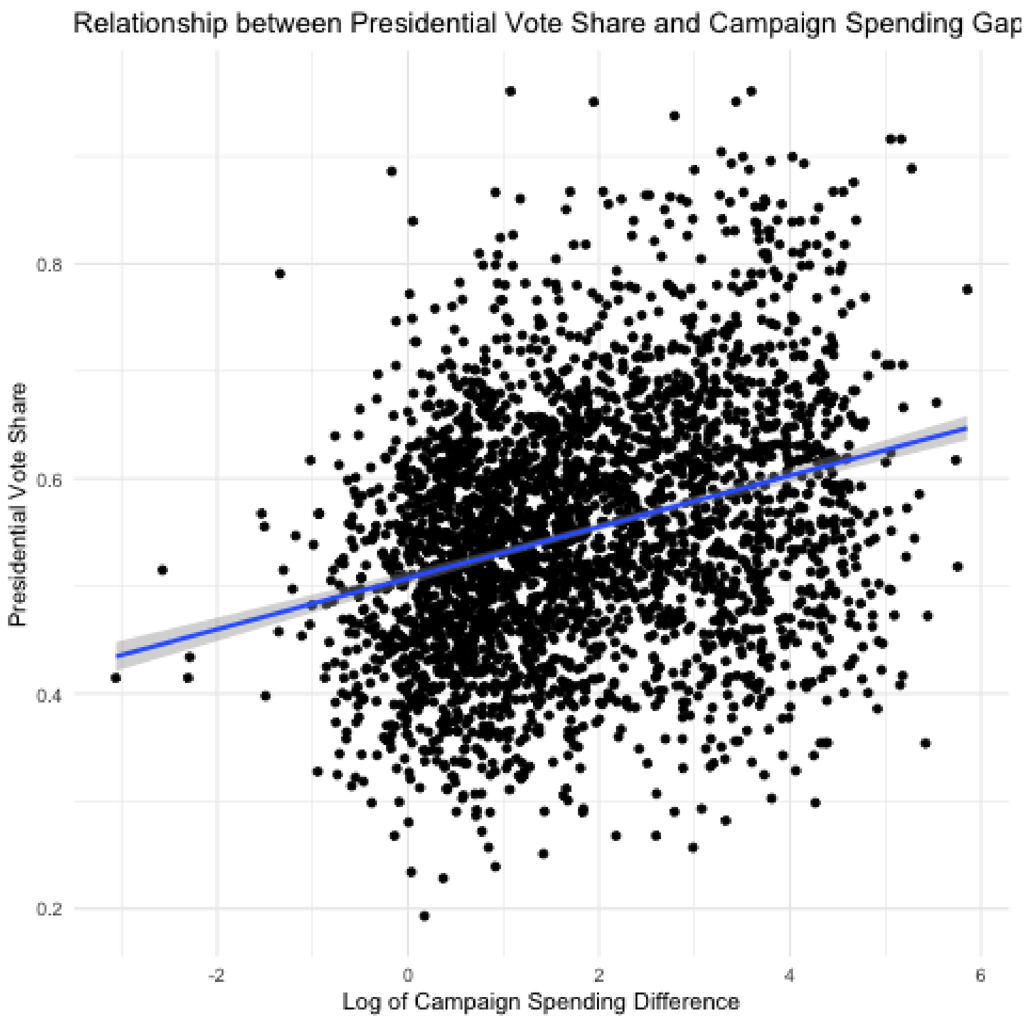
```r
fit2 <- lm(presvote ~ difflog, data = inc.sub)
fit2
stargazer(fit2)
```

Table 2:

	Dependent variable:
	Presidential Vote Share
Campaign Spending Difference	0.024***
	(0.001)
Constant	0.508***
	(0.003)
Note:	*p<0.1; **p<0.05; ***p<0.01

2. Make a scatterplot of the two variables and add the regression line.

Figure 2: scatterplot of difflog and presvote



3. Save the residuals of the model in a separate object.

```
# save residuals
resid_q2 <- fit2$residuals
resid_q2
```

4. Write the prediction equation.

$$\widehat{\text{PresVote}} = \hat{\alpha} + \hat{\beta} \times \text{DiffLog}$$
$$\widehat{\text{PresVote}} = 0.508 + 0.024 \times \text{DiffLog}$$

The intercept $\hat{\alpha} = 0.508$, represents the expected presidential vote share when the difference in campaign spending between incumbent and challenger is 0 (when both spend the same amount). The slope $\hat{\beta} = 0.024$ shows that for every one-unit increase in campaign spending difference, the presidential vote share increases by 0.024, on average.

Question 3

We are interested in knowing how the vote share of the presidential candidate of the incumbent's party is associated with the incumbent's electoral success.

1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `presvote`.

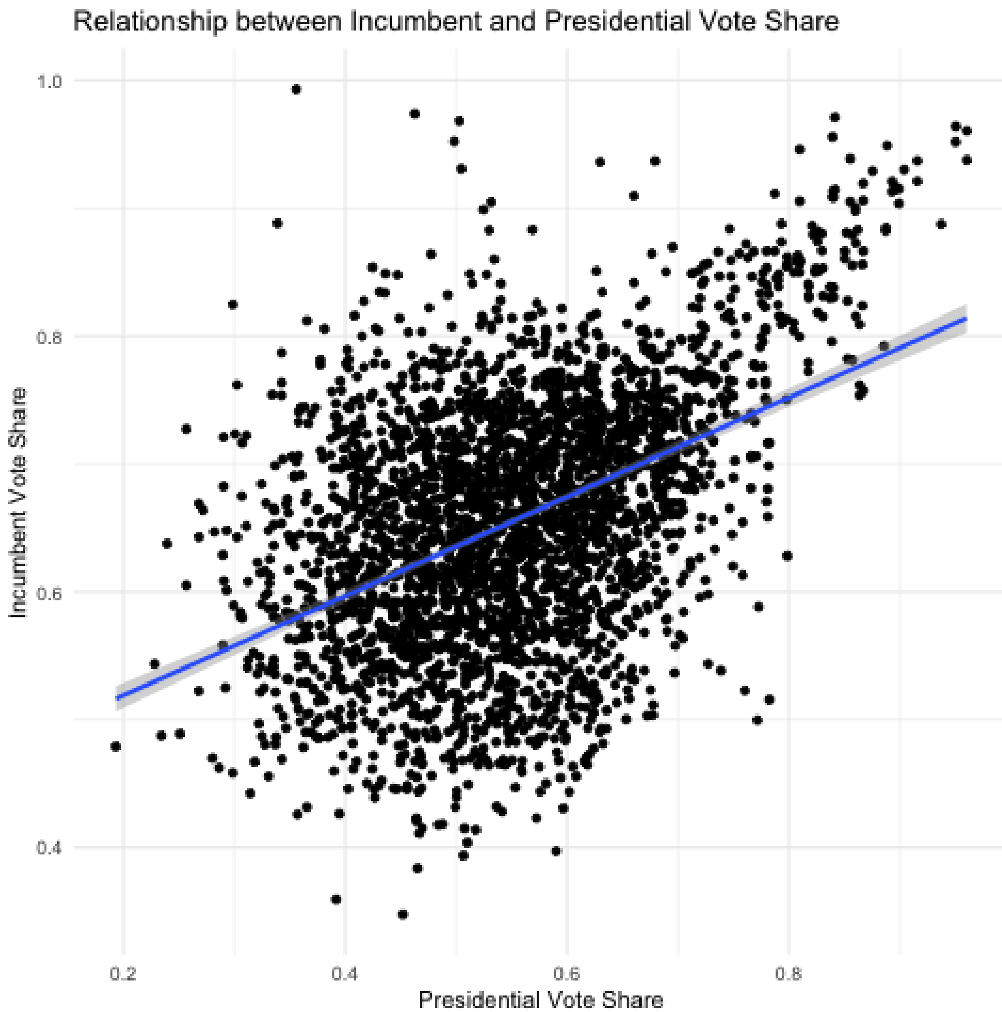
```
fit3 <- lm(voteshare ~ presvote, data = inc.sub)
fit3
stargazer(fit3)
```

Table 3:

	Dependent variable:
	Incumbent Vote Share
Presidential Vote Share	0.388^{***}
	(0.013)
Constant	0.441^{***}
	(0.008)
Note:	*p<0.1; **p<0.05; ***p<0.01

2. Make a scatterplot of the two variables and add the regression line.

Figure 3: Scatterplot of presvote and voteshare



3. Write the prediction equation.

$$\widehat{\text{VoteShare}} = \hat{\alpha} + \hat{\beta} \times \text{PresVote}$$

$$\widehat{\text{VoteShare}} = 0.441 + 0.388 \times \text{PresVote}$$

The intercept $\hat{\alpha} = 0.441$, represents the expected incumbent vote share when the ipresidential vote share is 0. Nevertheless, since a presidential vote share of 0 is not a realistic result, the value of the intercept has little to no substantive meaning. The slope $\hat{\beta} = 0.388$ shows that for every one-unit increase in presidential vote share, the incumbent vote share increases by 0.388, on average.

Question 4

The residuals from part (a) tell us how much of the variation in `voteshare` is *not* explained by the difference in spending between incumbent and challenger. The residuals in part (b) tell us how much of the variation in `presvote` is *not* explained by the difference in spending between incumbent and challenger in the district.

1. Run a regression where the outcome variable is the residuals from Question 1 and the explanatory variable is the residuals from Question 2.

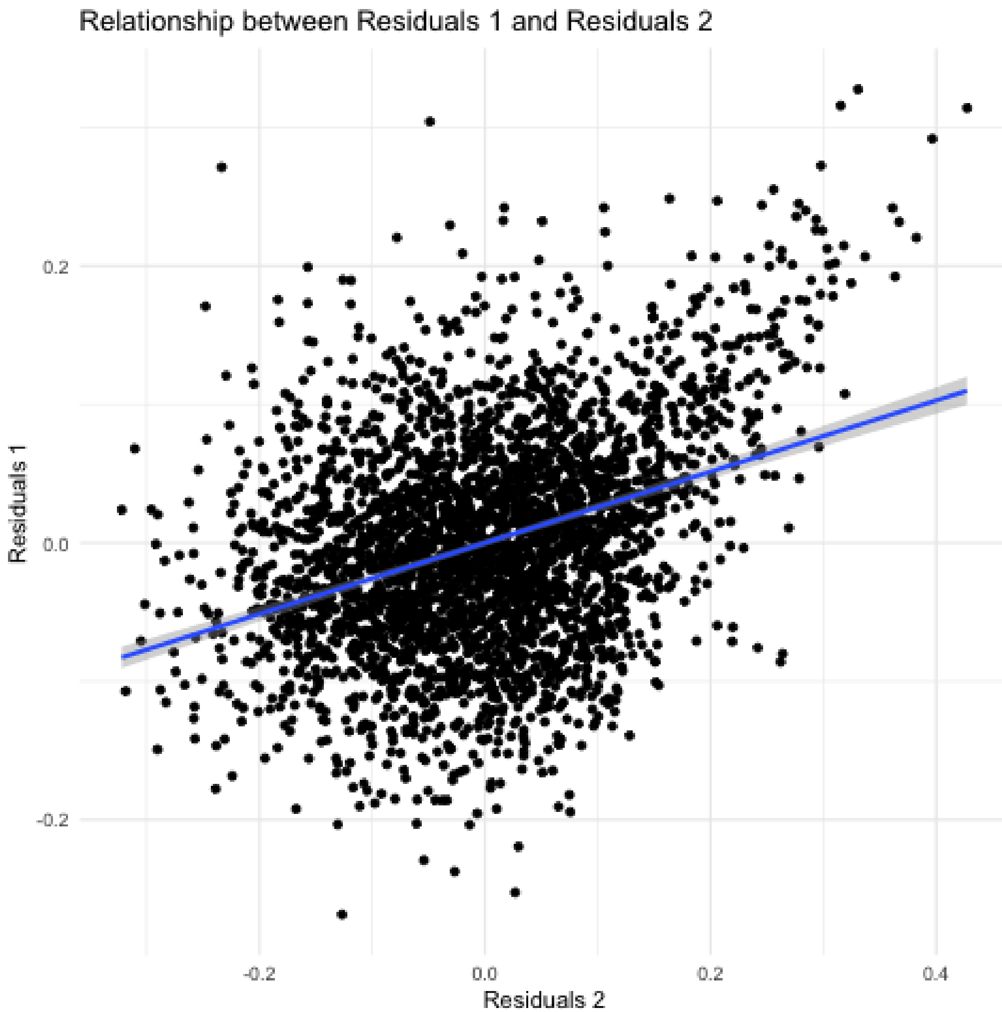
```r
# make a residuals data frame
resid_df <- data.frame(resid_q1, resid_q2)
resid_df

fit4 <- lm(resid_q1 ~ resid_q2, data = resid_df)
fit4
stargazer(fit4)
```

Table 4:

	Dependent variable:
	Residuals 1
Residuals 2	0.257***
	(0.012)
Constant	−0.000
	(0.001)
Note:	*p<0.1; **p<0.05; ***p<0.01

2. Make a scatterplot of the two residuals and add the regression line.

Figure 4: scatterplot of residuals 1 and residuals 2



3. Write the prediction equation.

$$\widehat{\text{Residuals1}} = \hat{\alpha} + \hat{\beta} \times \text{Residuals2}$$

$$\widehat{\text{Residuals1}} = 0 + 0.257 \times \text{Residuals2}$$

$$\widehat{\text{Residuals1}} = 0.257 \times \text{Residuals2}$$

The intercept $\hat{\alpha} = 0$, indicates that when Residuals2 $= 0$ the expected value of Residuals1 is also 0. The slope $\hat{\beta} = 0.257$ shows that for every one-unit increase in Residuals2, the value of Residuals1 increases by 0.257, on average.

Question 5

What if the incumbent's vote share is affected by both the president's popularity and the difference in spending between incumbent and challenger?

1. Run a regression where the outcome variable is the incumbent's `voteshare` and the explanatory variables are `difflog` and `presvote`.

```r
fit5 <- lm(voteshare ~ difflog + presvote, data = inc.sub)
fit5
stargazer(fit5)
```

Table 5:

	Dependent variable:
	Incumbent Vote Share
Campaign Spending Difference	0.036^{***}
	(0.001)
Presidential Vote Share	0.257^{***}
	(0.012)
Constant	0.449^{***}
	(0.006)
Note:	*p<0.1; **p<0.05; ***p<0.01

2. Write the prediction equation.

$$\widehat{\text{VoteShare}} = \hat{\alpha} + \hat{\beta}_1 \times \text{DiffLog} + \hat{\beta}_2 \times \text{PresVote}$$
$$\widehat{\text{VoteShare}} = 0.449 + 0.036 \times \text{DiffLog} + 0.257 \times \text{PresVote}$$

The intercept $\hat{\alpha} = 0.449$, represents the expected incumbent vote share when both the difference in campaign spending between incumbent and challenger and the presidential vote share are 0. The slope $\hat{\beta}_1 = 0.036$ shows that for every one-unit increase in campaign spending difference, the incumbent vote share increases by 0.036, on average, holding everything else constant. The slope $\hat{\beta}_2 = 0.257$ indicates that for every one-unit increase in presidential vote share, the incumbent vote share increases by 0.257, on average, holding everything else constant.

3. What is it in this output that is identical to the output in Question 4? Why do you think this is the case?

   The coefficient (and standard error) of the presidential vote share (`presvote`) in this multivariate regression is identical to the coefficient (and standard error) obtained by regressing `Residuals1` on `Residuals2`. This can be explained by understanding what each of these elements represents. Specifically, `Residuals1` show how much variation in `voteshare` is unexplained by `difflog`, while `Residuals2` indicate how much variation in `presvote` is not explained by `difflog`.

   Regressing `Residuals1` on `Residuals2`, we obtain the unconfounded effect of `presvote` on `voteshare`, after removing the effect of `difflog` from both variables. Within the multivariate regression, the coefficient of `presvote` also represents the effect of this variable on `voteshare`, while controlling for `difflog`, as its effect is accounted for separately. Thus, the coefficients of `presvote` and `Residuals2` are identical because they describe the same association within the data.